

DO YOU KYOTO?

Discover Sustainable Kyoto with SUTD

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Where does this come from? どういうこと?

Do You Kyoto?

A global catchphrase coined when Angela Merkel visited Kyoto in 2007 — meaning "Are you doing something good for the environment?" Kyoto City has promoted this since 2008, designating the 16th of every month as DO YOU KYOTO? Day.

「DO YOU KYOTO?」は、2007年にアンゲラ・メルケル氏が京都を訪問した際に生まれた、世界に向けた環境キャッチフレーズです。

この言葉には、「環境に良いことをしていますか?」という意味が込められています。京都市は2008年度からこれを環境行動のスローガンとして推進しており、京都議定書が発効した日である2005年2月16日にちなみ、毎月16日を「DO YOU KYOTO? デー」と定めています。



Kyoto has sustainable options — tourists just don't receive the right guidance at the moment they need it.

Tourists often make decisions based on convenience and familiarity. The problem is information timing, not intention.

「京都には持続可能な選択肢がある 必要な瞬間の正しい情報が届いていないだけ」



Hotspots overcrowded, alternatives underused

Gion, Nishiki Market, Arashiyama — heavily concentrated in the afternoon. Quieter routes, side streets, and temples nearby were nearly empty.



Sustainability practices not reaching tourists

Sustaina Kyoto and Miyako Ecology Center are resident-facing. Visitors rarely encounter sustainability guidance unless they deliberately seek it out.



Payment and transit rules confuse visitors

IC cards didn't always work across routes. Premium seats required extra payment tourists weren't aware of — friction that pushes people toward taxis.



Waste disposal not visible when needed

Cans and bottles left in restrooms — not from carelessness, but because tourists couldn't locate appropriate disposal points at the moment of need.

What we saw on the ground in Kyoto

「京都の現場で私たちが見たこと」

Tourist behaviour was not always irresponsible by intention — often, visitors lacked timely local guidance.

Field Observation	Reframed Insight	App Opportunity
Gion, Nishiki Market, and Arashiyama were crowded; nearby routes and alternatives were quietly underused.	Tourists follow default hotspots, not the best route.	Recommend scenic, lower-crowd alternatives nearby.
Sustaina Kyoto and Miyako Ecology Center seemed more local-facing than tourist-facing.	Sustainability knowledge exists, but tourists may not encounter it.	Translate local practices into tourist-friendly route nudges.
IC cards and premium-seat rules caused confusion; some cards failed across routes.	Public transport is sustainable, but not always intuitive for foreigners.	Add payment, IC-card, and seat guidance before boarding.
Bottles and cans left in restrooms — disposal points were hard to locate when needed.	Waste disposal fails when bins are hard to find at the moment of need.	Show nearby disposal/recycling points without adding new bins.

Problem Statement: How might we help tourists choose lower-carbon, less-crowded, and more respectful routes through Kyoto, while enabling locals to shape the guidance that protects their communities?

観光客が京都で低炭素で混雑を避け、地域に配慮したルートを選べるようにしながら、地域住民が自分たちのコミュニティを守るための案内づくりに参加できるようにするにはどうすればよいだろうか？

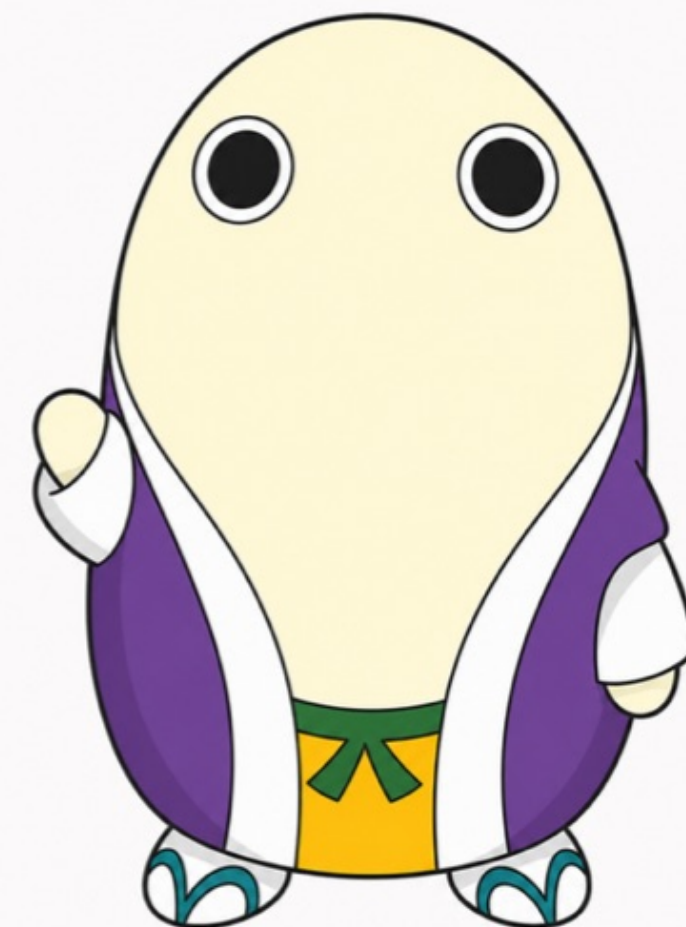
DO YOU

京途?

A sustainable route companion for responsible Kyoto travel

Our twist: 京都 (Kyoto the city) becomes 京途 (Kyoto the journey). We ask not just whether you care about the environment — but HOW you move through Kyoto. The route you choose is an environmental act.

私たちのコンセプトは、「京都」を「京途」へと捉え直すことです。京都という場所を訪れるだけでなく、どのような「途」をたどるのかに目を向けます。大切なのは、環境を意識しているかどうかだけではありません。京都の中をどう移動するか。その一つひとつのルート選択が、環境へのアクションになるのです。



Mayumaro まゆまる · Kyoto's mascot

「京途を歩こう！」
"Let's walk the Kyoto route!"

Tourist Guidance, Local Stewardship

観光客への案内、地域住民によるまちの見守り



DO YOU 京途? is not just a tourist chatbot. It is a local-informed route companion where tourists receive better guidance, while locals shape what "better" means.

TOURISTS RECEIVE

- Lower-carbon route options
- Less-crowded alternatives
- Etiquette tips before arrival
- Fare, IC-card & transit guidance
- Bin / carry-trash guidance

LOCALS CONTRIBUTE

- Crowd hotspot reports
- Sensitive / residential zones
- Local manners & seasonal updates
- Practical transport advice
- Bin and recycling point updates

Tourists choose better routes. Locals shape the guidance that protects Kyoto.

観光客はより良いルートを選び、地域住民は京都を守るための案内づくりに参加する。

Mayumaro guides you. You choose how.

「まゆまろが案内します。選ぶのはあなた。」



景 SCENIC ROUTE

For the explorer. Slightly longer route that passes through highly-rated spots — temples, gardens, viewpoints — along the way. Still fully on public transport or walking.

Prioritises:

Scenic Value • Cultural depth • Crowd-aware timing

Google Maps data + time-of-day patterns + community reports → crowd-aware routing

急 QUICK ROUTE

For the efficient. Fastest point-to-point via public transport. Minimise transfers, minimise time, still zero private car.

Prioritises:

Carbon-Light • Fastest public transit • Fewer transfers, lower carbon

Google Directions API → transit departure/arrival scheduling

How scores are actually computed

「スコアの実際の計算方法」



Carbon-Light

Computed from Directions API transport legs. Walking = 10, Train/Subway = 8, Bus = 6, Taxi = 2. Weighted average by leg distance.

 *Transport mode + route distances*



Congestion-Light

Google Places Popular Times for the destination. If currently BUSY → suggests nearby alternatives with high ratings (≥ 4.0 ★) and lower footfall instead.

 *Time-of-day patterns + community reports + available crowd proxies*



Scenic Value

Places API fetches cultural POIs (temples, parks, viewpoints) along the route. Count and average rating drive the score. Used mainly in Scenic mode.

 *Cultural POIs along/near route*



Etiquette Readiness

Positive score: has the chatbot surfaced the relevant local travel tips for this route? Triggered by transport mode, location type, and time of day from curated knowledge base.



Function + Livability + Purpose



FUNCTION

Performs effectively

- Accurate routes via Google Directions API with live traffic
- Natural language queries — no map literacy or app download needed
- Scenic vs Quick mode — user chooses their travel intention
- 4 sustainability scores computed from real API data + curated rules



LIVABILITY

Enhances human, cultural & environmental well-being

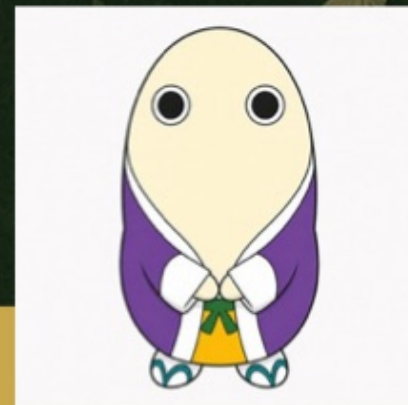
- Diverts tourist flow from crowded hotspots to rated alternatives nearby
- Promotes walking and public transit — reduces congestion and emissions
- Contextual etiquette tips improve tourist-resident harmony in real time
- Encourages slower, scenic travel that spreads footfall across the city



PURPOSE

Addresses meaningful need & creates positive impact

- Aligned with Kyoto City's DO YOU KYOTO? environmental campaign
- Directly supports Resilient Kyoto Strategy Initiative 4-3-2
- Nudges tourist behaviour at the moment of decision — most impactful point
- Scalable to any heritage city facing over-tourism and congestion



Mayumaro as your guide — with contextual etiquette tips

Mayumaro shows route-specific boarding and payment instructions before you board.



まゆまる ・DO YOU 京途?

I want to go to Fushimi Inari from Kyoto Station, its 3pm rn. Help!

⚠ Fushimi Inari is BUSY right now (3pm peak). Mayumaro suggests:

Tofuku-ji Temple

8 min walk • 4.4★ • Quiet now

Explore quieter Fushimi Momoyama Castle alternatives

Sake district / canal area / Momoyama Castle Park by transit

Etiquette Tips for This Route



Subway etiquette

Stand on the left on escalators. Queue behind the yellow line. No calls on trains.

Fushimi Inari

Keep to the side paths. Speak quietly. This is a sacred site, not just a photo spot.

City bus

Board from the rear, tap IC card on entry and exit. Priority seats near the driver.

Rubbish

Very few public bins in Kyoto — carry your trash back with you until you find one.

The knowledge gap that sends tourists to taxis

「観光客をタクシーへ向かわせる情報の格差」

Fare Comparison

Kyoto Sta. → Arashiyama

 ¥240 (Sagano Line)

 ¥2,500–3,500

Gion → Kinkaku-ji

 ¥230 (City Bus 205)

 ¥1,400–1,800

Kyoto Sta. → Fushimi Inari

 ¥150 (JR Nara Line)

 ¥1,200–1,600

How to Actually Pay

Flat-fare bus (City Bus)

¥230 flat. Board rear. Tap IC or pay cash on EXIT only. No ticket.

Distance-based bus (Kyoto Bus)

Take numbered ticket on ENTRY. Pay by zone on EXIT. IC taps both entry & exit.

IC cards (Suica / ICOCA)

Interchangeable on City Bus & subway. Patchy on Hankyu, Randen, some Kyoto Bus routes. Some rural stations: pay driver.

8+ operators, no unified system

Keihan, Hankyu, Kintetsu, JR, Eizan, Keifuku — all separate fare rules and IC policies. Sagano Romantic Train: cash only.

End-to-end example: Kyoto Station → Arashiyama

「使用例：京都駅から嵐山へ」



"Kyoto Station to Arashiyama, 3pm, Scenic please"



まゆまる recommends (Scenic 景):

Route: Sagano Line from Kyoto Station → Saga-Arashiyama. Stop en route at Nijo Castle (4.2★, 8 min walk off route) — quiet at 3pm.

Score: Carbon-Light 9/10 • Congestion-Light 7/10 • Scenic Value 8/10 • Etiquette Readiness ✓

Timing: Arashiyama busy 2–4pm. Arriving ~3:30pm is borderline — consider Tenryu-ji Gardens (less crowded alternative nearby, 4.5★).

Etiquette: ⚠ Sagano Line escalators: stand left. At bamboo grove: no drones, keep voices low near temple grounds.

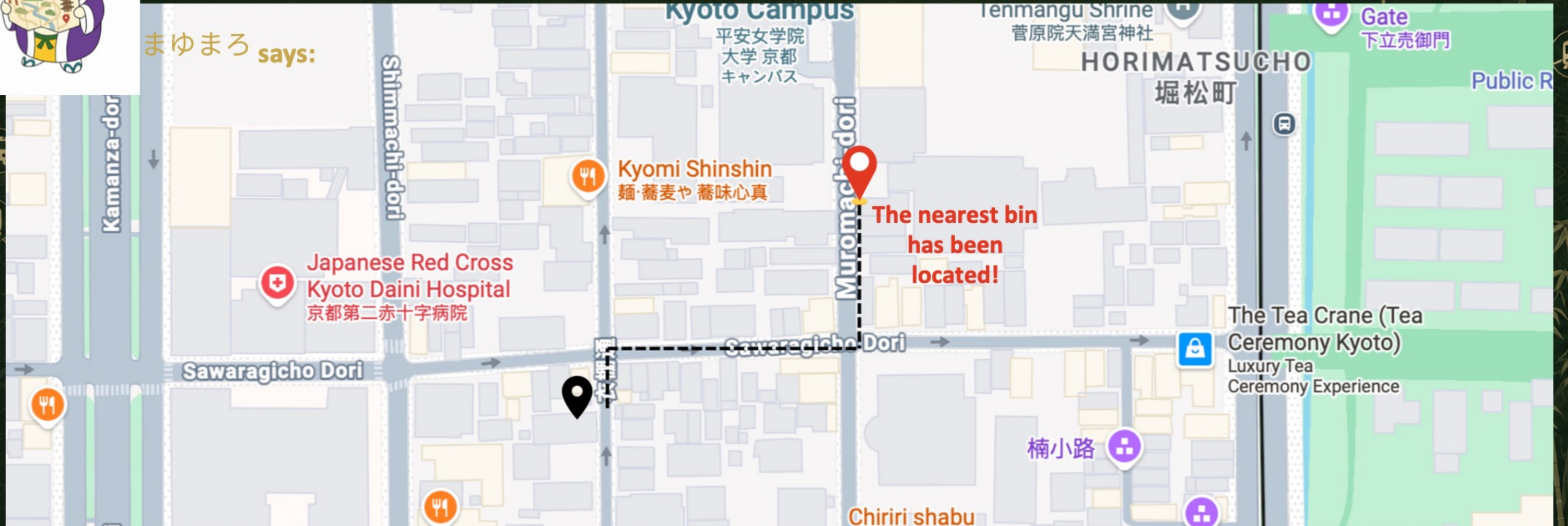
Bin Locating

「近くのごみ箱を探す」



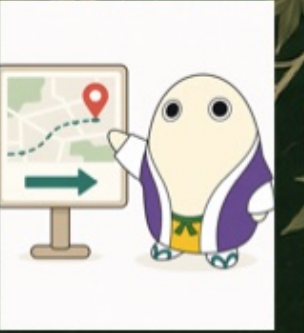
まゆまろ says:

🗑️ "Help! Where can I throw my trash!"



Not Every Hidden Gem Should Be Exposed

すべての穴場を紹介するべきではない



A sustainable tourism app should protect quiet places, not turn them into the next overcrowded hotspot.

LOCAL SETTING

TOURIST-FACING ACTION

Recommend openly

Show as an alternative route or stop

Recommend only at quiet times

Show only during off-peak hours

Etiquette-only zone

Do not promote; only show behaviour tips

Sensitive / avoid zone

Route tourists away where appropriate

Local knowledge should protect places, not exploit them.

地域の知識は、場所を消費するためではなく、守るために使われるべきである。

Kyoto City's own strategy backs this project

「京都市自身の戦略がこのプロジェクトを後押し」



Sustainable
Cities



Climate
Action



Responsible
Consumption



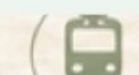
Partnerships
for Goals

DO YOU KYOTO? — Kyoto City, 2008

Car modal share dropped from 28.3% to 22.1% between 2000–2015. 1 in 5 car users switched to trains or buses. Our chatbot extends this behavioural shift to tourists.

Resilient Kyoto Strategy — Initiative 4-3-2

"Public Transport Promotion Campaign — engage communities, businesses and universities to improve and promote use of Kyoto's public transport (train and bus). Reduces traffic congestion, greenhouse gas emissions, and improves air quality."



Resilient Kyoto Strategy — Initiative 4-3-1

"Resilience of Kyoto's Pedestrian Network — improve walkability and pedestrian access to public transport stations." Our Scenic routing suggestions directly support this by surfacing walkable alternatives.

How We Measure Impact

「効果測定の方法」

Impact area	Possible metric
Lower-carbon travel	Estimated taxi/private-car trips avoided
CO ₂ reduction	Estimated emissions saved per recommended route
Crowd dispersion	Number of users redirected from crowded hotspots to alternative routes
Respectful behaviour	Etiquette tips shown before arrival at sensitive areas
Local participation	Verified local contributions: bins, crowd reports, etiquette updates, avoid zones
Tourist usability	% of users choosing Scenic / Quick route after seeing the recommendation

移動の瞬間にインパクトを測る
観光客が混雑した定番ルートではなく、低炭素で、混雑を避け、地域に配慮したルートを選んだかを評価する。

How it's built

「システムの構築方法」

01 RAG Knowledge Base

Curated field notes + Kyoto policy docs + community contributions. Retrieved per query — not generic AI.

02 Google Maps APIs

Directions API (live traffic, legs) + Places API (popular times, POI ratings, nearby search).

03 Claude AI Layer

API scores + retrieved knowledge injected into system prompt. Claude writes narrative, Mayumaro delivers.

04 Streamlit Frontend

Chat UI with Scenic/Quick toggle, score cards, and contextual etiquette pop-up.

05 Two Versions

Tourist (EN) reads from knowledge base. Local (JP) contributes to it — crowd data, bins, hidden gems.

Community Knowledge Layer

コミュニティの知識レイヤー

The RAG knowledge base has two layers: (1) Curated — field notes, policy docs, verified transit rules the team validates. (2) Community — locals and returning visitors contribute bin locations, crowd patterns, seasonal etiquette updates, and hidden gem spots. Both layers are retrieved per query to ground Claude's responses in hyperlocal reality no generic AI model has.

Role delegation

Chatbot Backend

Anthropic API, Google Maps integration, scoring logic pipeline

Transport & Data

Route logic, public transport data, curated etiquette knowledge base

Frontend & UX

Streamlit UI, Mayumaro integration, Scenic/Quick toggle, etiquette pop-up

Research & Deck

Kyoto policy research, trip documentation, presentation and final report

Why Locals Would Use It

地域住民が参加する理由

Locals are not just helping tourists — they are protecting their own daily environment.



01 Reduce pressure on daily life

住宅街・バス停・生活道路での混雑、ごみ、騒音、迷惑行為を減らす。

02 Guide tourists before problems happen

住民が直接注意しなくても、まゆまるが事前にやさしくマナーを伝える。

03 Support small local businesses

小さなお店、カフェ、工芸店、商店街へ人の流れを分散する。

04 Protect sensitive places

「おすすめ」「静かな時間帯のみ」「紹介しない」など、案内の強さを調整できる。

Design implication: local participation must feel protective, not extractive.

From Local Input to Tourist Nudge

地域の声から観光客へのナッジへ

The system turns local knowledge into route-level micro-guidance at the moment of decision.



1

Local Input

Residents, shop owners and returning visitors submit updates.

2

Verification

Team / community moderators check usefulness, safety and respectfulness.

3

Route Logic

Knowledge is linked to place, time, mode and route context.

4

Tourist Nudge

Mayumaro delivers a simple suggestion before the route is chosen.

Example tourist-facing output:

"This street is crowded and residential. Mayumaro suggests walking via the riverside path instead — 4 minutes longer, quieter, and more scenic."

「この通りは住宅街で混雑しやすいため、川沿いの道をおすすめします。所要時間は4分長くなりますが、より静かで景色の良いルートです。」